WMMSE演算法

Weighted minimum mean square error

$$y_{ik} = H_{ik} V_{ik} s_{ik} + \sum_{m=1, m\neq i}^{I_k} H_{ik,k} V_{mk} s_{mk} + \sum_{j\neq k, j=1}^{K} \sum_{l=1}^{I_j} H_{ikj} V_{lj} s_{lj} + n_{ik}$$

$s_{ik} \in \mathbb{C}^{d_{ik}\times 1}$: k-th BS 傳給 i_k 個 UE

$$E[s_{ik} s_{ik}^H] = I$$

共 I 個 UE
K 個 BS

$V_{ik} \in \mathbb{C}^{M_k \times d_{ik}}$: 波束成形矩陣

$H_{ikj} \in \mathbb{C}^{N_{ik}\times M_j}$: j-th BS 到 i_k 個 UE 的通道

$n_{ik} \in \mathbb{C}^{N_{ik}\times 1} \sim CN(0, \sigma_{ik}^2 I)$ 高斯白雜訊

y_{ik} : 用戶 i_k 的接收信號

Estimate signal

$$\hat{s}_{ik} = U_{ik}^H y_{ik}$$

Model :

$$\max \sum_{k=1}^{K} \sum_{i_k=1}^{I_k} \alpha_{ik} R_{ik}$$

$$\text{s.t.} \sum_{i=1}^{I_k} Tr(V_{ik} V_{ik}^H) \leq P_k \ , \ \forall k = 1, 2, \ldots, K$$

$$R_{ik} \triangleq \log\det\left(I + H_{ik,k} V_{ik} V_{ik}^H H_{ik,k}^H \left(\sum_{(l,j)\neq(i,k)} H_{ikj} V_{lj} V_{lj}^H H_{ikj}^H + \sigma^2 I\right)^{-1}\right)$$

用F_{ik}接收信號與真實信號之間MSE形式可表示為

$$E_{ik} \triangleq E_{sn}\left[(\hat{S}_{ik}-S_{ik})(\hat{S}_{ik}-S_{ik})^H\right]$$

$$= E_{sn}\left[(U_{ik}^H y_{ik} - S_{ik})(y_{ik}^H U - S_{ik}^H)\right]$$

$$= E_{sn}\left[U_{ik}^H y_{ik} y_{ik}^H U_{ik} - U_{ik}^H y_{ik} S_{ik}^H - S_{ik} y_{ik}^H U_{ik} + S_{ik} S_{ik}^H\right]$$

$$= E_{sn}\Big[U_{ik}^H\Big(H_{ik}V_{ik}S_{ik} + \sum_{(k,j)\neq(i,k)} H_{ik}V_{lj}S_{lj} + n_{ik}\Big)\Big(S_{ik}^H V_{ik}^H H_{ik}^H + \sum_{(k,j)\neq(i,k)} S_{lj}^H V_{lj}^H H_{ikj}^H + n_{ik}^H\Big)U_{ik}$$

$$- U_{ik}^H\Big(H_{ik}V_{ik}S_{ik} + \sum_{(k,j)\neq(i,k)} H_{ikj}V_{lj}S_{lj} + n_{ik}\Big)S_{ik}^H - S_{ik}\Big(S_{ik}^H V_{ik}^H H_{ik}^H + \sum_{(k,j)\neq(i,k)} S_{lj}^H V_{lj}^H H_{ikj}^H + n_{ik}^H\Big)U_{ik}$$

$$+ S_{ik}S_{ik}^H\Big]$$

$$= I - U_{ik}^H H_{ik} V_{ik} - V_{ik}^H H_{ik}^H U_{ik} + \sum_{(l,j)} U_{ik}^H H_{ikj} V_{lj} V_{lj}^H H_{ikj}^H U_{ik} + \sigma_{ik}^2 U_{ik}^H U_{ik}$$

基於MSE建構的波束成形的優化問題

$$\min_{U,V} \sum_{k=1}^{K}\sum_{i=1}^{I_k} Tr(E_{ik})$$

$$s.t. \quad \sum_{i=1}^{I_k} Tr(V_{ik}V_{ik}^H) \le P_k, \quad k=1,2,\dots,K$$

拉格朗日：

$$\mathcal{L}(U,V,\lambda) = \sum_{k=1}^{K}\sum_{i=1}^{I_k} Tr(E_{ik}) + \sum_{k=1}^{K}\lambda_k\Big(\sum_{i=1}^{I_k} Tr(V_{ik}V_{ik}^H) - P_k\Big)$$

$$\frac{\partial \mathcal{L}(U,V,\lambda)}{\partial U_{ik}} = -2H_{ik}V_{ik} + 2\sum_{j=1}^{K}\sum_{l=1}^{I_j} H_{ikj}V_{lj}V_{lj}^H H_{ikj}^H U_{ik} + 2\sigma_{ik}^2 U_{ik} = 0$$

MMSE 接收器公式

$$U_{ik}^{mmse} = J_{ik}^{-1} H_{ikk} V_{ik}$$

$$J_{ik} \triangleq \sum_{k=1}^{K} \sum_{l=1}^{I_j} H_{ikj} V_{lj} V_{lj}^H H_{ikj}^H + \sigma_{ik}^2 I$$

代回 E_{ix} 式

$$E_{ik}^{mmse} = I - V_{ik}^H H_{ikk}^H \left(2J_{ik}^{-1} - J_{ik}^{-1} \sum_{j=1}^{K} \sum_{l=1}^{I_j} H_{ikj} V_{lj} V_{lj}^H H_{ikj}^H J_{ik}^{-1} - \sigma_{ik}^2 J_{ik}^{-1} J_{ik}^{-1} \right) H_{ik} V_{ik}$$

$$= I - V_{ik}^H H_{ikk}^H J_{ik}^{-1} H_{ikk} V_{ik}$$

WMMSE

$$\min_{W,U,V} \sum_{k=1}^{K} \sum_{i=1}^{I_k} \alpha_{ik} \left(Tr(W_{ik} E_{ik}) - \log\det(W_{ik}) \right)$$

$$s.t. \quad \sum_{i=1}^{I_k} Tr(V_{ik} V_{ik}^H) \le P_k, \quad k = 1, 2, \dots K$$

固定 U,V 的话，问题是 convex.

$$\sum_{k=1}^{K} \sum_{i=1}^{I_k} \alpha_{ik} \left(Tr(W_{ik} E_{ik}) - \log\det(W_{ik}) \right) + \lambda \left(\sum_{i=1}^{I_k} Tr(V_{ik} V_{ik}^H) - P_k \right) = 0$$

$$\xrightarrow[\frac{\partial}{\partial W_{ik}}]{} \alpha_{ik} E_{ik}^H - \alpha_{ik} (W_{ik}^{-1})^H = 0$$

$$\longrightarrow W_{ik}^{opt} = E_{ik}^{-1}$$

代回

$$\min_{V} \sum_k \sum_i \cancel{\alpha_{ik} I} - \log\det\left((E_{ik}^{mmse})^{-1} \right)$$

可省

$$\rightarrow \max_{V} \sum \sum \log\det\left((E_{ik}^{mmse})^{-1} \right)$$

$$\max_{V} \sum_{k=1}^{K}\sum_{i=1}^{I_k} \alpha_{ik} \log\det\left(\left(E_{ik}^{mmse}\right)^{-1}\right)$$

$$\text{s.t. } Tr\left(V_{ik}V_{ik}^H\right) \le P_k,\ \forall i_k \in \mathcal{I}$$

利用伍德值林矩阵恒等式

$$(A+UCV)^{-1} = A^{-1} - A^{-1}U\left(C^{-1}+VA^{-1}U\right)^{-1}VA^{-1}$$

令 $A=I$, $U=-V_{ik}^H H_{ik}^H$, $C=J_{ik}^{-1}$, $V=H_{ikk}V_{ik}$

$$\log\det\left(\left(E_{ik}^{mmse}\right)^{-1}\right) = \log\det\left(I+V_{ik}^H H_{ikk}^H \Gamma_{ik}^{-1} H_{ikk} V_{ik}\right)$$

$$= \log\det\left(I+H_{ikk}V_{ik}V_{ik}^H H_{ikk}^H \Gamma_{ik}^{-1}\right)$$

相等

其中, $\Gamma_{ik} \triangleq \sum_{(l,j)\neq(i,k)} H_{ik,j} V_{lj} V_{lj}^H H_{ik,j}^H + \sigma_{ik}^2 I$

已知 $R_{ik} \triangleq \log\det\left(I+H_{ikk}V_{ik}V_{ik}^H H_{ikk}^H\left(\sum_{(l,j)\neq(i,k)} H_{ik,j}V_{lj}V_{lj}^H H_{ik,j}^H + \sigma_{ik}^2 I\right)^{-1}\right)$

得證

當用BCD(塊狀梯度下降法時), 依次代入最佳值(W和U都有closed form), Precoder 矩陣V的優化在兩個問題上等價, 有相同的最佳解

Conclusion:

TABLE I

PSEUDO CODE OF THE PROPOSED WMMSE ALGORITHM FOR MIMO-IBC

Initialize $\mathbf{V}_{i_k}$'s such that $\mathrm{Tr}\left(\mathbf{V}_{i_k}\mathbf{V}_{i_k}^H\right) = \frac{p_k}{I_k}$

repeat

$\mathbf{W}'_{i_k} \leftarrow \mathbf{W}_{i_k}, \quad \forall\, i_k \in \mathcal{I}$

$\mathbf{U}_{i_k} \leftarrow \left(\sum_{(j,\ell)} \mathbf{H}_{i_k j}\mathbf{V}_{\ell_j}\mathbf{V}_{\ell_j}^H\mathbf{H}_{i_k j}^H + \sigma_{i_k}^2\mathbf{I}\right)^{-1}\mathbf{H}_{i_k k}\mathbf{V}_k, \ \forall\, i_k \in \mathcal{I}$

$\mathbf{W}_{i_k} \leftarrow \left(\mathbf{I} - \mathbf{U}_{i_k}^H\mathbf{H}_{i_k k}\mathbf{V}_{i_k}\right)^{-1}, \ \forall\, i_k \in \mathcal{I}$

$\mathbf{V}_{i_k} \leftarrow \alpha_{i_k}\left(\sum_{(j,\ell)}\alpha_{\ell_j}\mathbf{H}_{\ell_j k}^H\mathbf{U}_{\ell_j}\mathbf{W}_{\ell_j}\mathbf{U}_{\ell_j}^H\mathbf{H}_{\ell_j k} + \mu_k^*\mathbf{I}\right)^{-1}\mathbf{H}_{i_k k}^H\mathbf{U}_{i_k}\mathbf{W}_{i_k}, \ \forall\, i_k$

until $\left|\sum_{(j,\ell)}\log\det\left(\mathbf{W}_{\ell_j}\right) - \sum_{(j,\ell)}\log\det\left(\mathbf{W}'_{\ell_j}\right)\right| \le \epsilon$

γ Covariance matrix

核心:

1° $\boxed{R = \log\det\left(E^{-1}\right)}$

2° MMSE receiver closed-form

$U = (\text{signal} + \text{interference} + \text{noise})^{-1}\,\text{signal}$

$$\boxed{= \left(\sum_{(j,\ell)} H_{ik}V_{\ell j}V_{\ell j}^H H_{ik}^H{}_j + \sigma_{ik}^2 I\right)^{-1} H_{ik\,k}V_k}$$

$= \gamma^{-1} H_{kk} V_k$

3° $W = E^{-1}$ closed-form $W = E^{-1} = (I - U^H H V)^{-1}$

由于用mmse receiver，也就是 $U = (\text{signal} + \text{interference} + \text{noise})^{-1}\,\text{signal}$

$$E_k = I - U_k^H H_{kk} V_k - V_k^H H_{kk}^H U_k + \underbrace{\sum_j U_k^H H_{kj} V_j V_j^H H_{kj}^H U_k}_{\text{干扰}} + \underbrace{\sigma^2 U_k^H U_k}_{\text{Noise}}$$

干扰 + noise

$$\longrightarrow \sum U_k^H H_{kj} V_j V_j^H H_{kj}^H U_k + U_k^H (\sigma^2 I) U_k$$

$$= U_k^H \left(\sum_j H_{kj} V_j V_j^H H_{kj}^H + \sigma^2 I \right) U_k$$

定义 $\Upsilon = \sum_j H_{kj} V_j V_j^H H_{kj}^H + \sigma^2 I$ 称 covariance matrix

$$= U_k^H \Upsilon U_k$$

消去 消去

$$E_k = I - U_k^H H_{kk} V_k \boxed{- V_k^H H_{kk}^H U_k} + \boxed{U_k^H \Upsilon U_k}$$

代入 MMSE receiver $U = \Upsilon^{-1} H_{kk} V_k$

$$V_k^H H_{kk}^H \Upsilon^{-1} \Upsilon \Upsilon^{-1} H_{kk} V_k$$

$$= V_k^H H_{kk}^H \Upsilon^{-1} H_{kk} V_k$$

$$\| $$

$$- V_k^H H_{kk}^H \Upsilon^{-1} H_{kk} V_k$$

$$\boxed{\begin{aligned} E_k &= I - U_k^H H_{kk} V_k \\ W &= E_k^{-1} = (I - U_k^H H_{kk} V_k)^{-1} \end{aligned}}$$

4° WMMSE 目標

$$\max \quad \alpha R = \alpha \log\det(I + SINR)$$

$$\Updownarrow$$

$$\max \ \alpha \log\det(E_k^{-1})$$

$$\Updownarrow$$

$$\min \ \alpha\left(Tr(W_k E_k) - \log\det(W_k)\right)$$

5° 演算法用 BCD 框架

$U \to W \to V$ 反覆更新

更新 V 這步 (更新 V 的子問題)

和 V 無關

$$\min_{V_k} \sum_k \alpha_k Tr(W_k E_k) - \cancel{\log\det(W_k)}$$

$$s.t. \quad Tr(V_k V_k^H) \le P_k$$

已知

$$E_k = I \boxed{- U_k^H H_{kk} V_k} \boxed{- V_k^H H_{kk}^H U_k} + \boxed{\sum_j U_k^H H_{kj} V_j V_j^H H_{kj}^H U_k} + \sigma^2 U_k^H U_k$$

代入 $\sum_k \alpha_k Tr(W_k E_k)$

乘上 W_k: $-\alpha_k Tr(W_k U_k^H H_{kk} V_k)$

乘上 W_k

$$-\alpha_k Tr(W_k V_k^H H_{kk}^H U_k)$$

乘上 W_k

$$\sum_j \alpha_j Tr(W_j U_j^H H_{jk} V_k V_k^H H_{jk}^H U_j)$$

$$\sum_j \alpha_j Tr(W_j U_j^H H_{jk} V_k V_k^H H_{jk}^H U_j) = \sum_j \alpha_j Tr(V_k^H H_{jk}^H U_j W_j U_j^H H_{jk} V_k)$$

由於 $W_k = E_k^{-1}$

$E_k = E[e_k e_k^H] \rightarrow$ Hermitian $\rightarrow W_k = W_k^H$

$$-\alpha_k Tr(W_k U_k^H H_{kk} V_k) = -\alpha_k Tr(W_k V_k^H H_{kk}^H U_k)$$

更新 V 的子問題

$$\min_{V_k} \sum_k \alpha_k Tr(W_k E_k)$$

$$s.t. \quad Tr(V_k V_k^H) \le P_k$$

$\Downarrow$

$A_k = \sum_j$ 紀对所有人的干擾影響　想把signal 對準紀的方向

$$\min Tr\Big(V_k^H \underbrace{\boxed{\Big(\sum_j \alpha_j H_{jk}^H U_j W_j U_j^H H_{jk}\Big)}}_{A_k} V_k\Big) - 2Re\Big(Tr\Big(V_k^H \underbrace{\boxed{\alpha_k H_{kk}^H U_k W_k}}_{B_k}\Big)\Big)$$

$$s.t. \quad Tr(V_k V_k^H) \le P_k$$

$$\Rightarrow \min_{V_k} Tr(V_k^H A_k V_k) - 2Re\{Tr(V_k^H B_k)\}$$

$$s.t. \quad Tr(V_k V_k^H) \le P_k$$

quadratic optimization

$$\mathcal{L}(V_k, \mu_k) = Tr(V_k^H A_k V_k) - 2Re\{Tr(V_k^H B_k)\} + \mu_k(Tr(V_k V_k^H) - \underbrace{P_k}_{\text{和 V 無關}})$$

對 V_k 微分後 $=0$

$$f(V_k) = Tr(V_k^H A_k V_k) - 2Re\{Tr(V_k^H B_k)\} + \mu_k Tr(U^H U)$$

$$= Tr(V_k^H \underbrace{(A_k + \mu_k I)}_{\triangleq C_k} V_k) - 2Re\{Tr(V_k^H B_k)\}$$

$$f(V_k) = Tr(V_k^H C_k V_k) - 2Re\{Tr(V_k^H B_k)\}$$

公式 $\frac{\partial}{\partial V} Tr(V^H C V) = CV$ (當 $C = C^H$)

$$\frac{\partial}{\partial V} Tr(V^H B) = B$$

$$\frac{\partial}{\partial V} Re\{Tr(V^H B)\} = B$$

$$\frac{\partial}{\partial V} f(V_k) = C_k V_k - B_k = 0$$

$$(A_k - \mu_k I) V_k = B_k \rightarrow \boxed{V_k = (A_k - \mu_k I)^{-1} B_k}$$

closed-form

μ 可由 Bisection 演算法求

WMMSE 到底在做什麼?

1° 把 non-convex rate problem 變成 weighted + MSE 的等價問題

$$\max \log\det(I + SINR) \longrightarrow \min Tr(WE) - \log\det(W)$$

2° 好處, 做 BCD, 固定另外兩個變數, 每一步都是 convex

收斂到 local optimal

3° 每一步都有 closed-form

每次對 WMMSE 更新 U, W, V 等價在做 rate problem 的

U, W, V 更新